Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
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Bhimtal Campus

Term-Work

Of

Compiler Design Lab (PCS-601)

Submitted in partial fulfilment of the requirement for the VI semester

Bachelor of Technology

(Computer Science & Engineering)

By

Mayank Bisht

2361323

Under the Guidance of

Mr.Devesh Pandey

Assistant Professor

Ms.Vaishali Dev

Teaching Assistant

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

GRAPHIC ERA HILL UNIVERSITY, BHIMTAL CAMPUS

SATTAL ROAD, P.O. BHOWALI

2025-2026

CERTIFICATE

The Term work of **Compiler Design Lab**, being submitted by Mayank Bisht **D/O** Mr.Heera Singh Bisht , Enrollment no 230112060, Roll no. 2361323 to **Graphic Era Hill University Bhimtal Campus** for the award of bona fide work carried out by her. She has worked under our guidance and supervision and fulfilled the requirement for the submission of this work report.

(……) **(……)**

Faculty-in-Charge **HOD, CSE Dept.**

ACKNOWLEDGEMENT

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STUDENT'S DECLARATION

I Mayank Bisht hereby declare the work, which is being presented in the term-work, entitled **Compiler Design Lab** in partial fulfillment of the requirement for the award of the degree **B.Tech** in the session **2025-2026**, is an authentic record of my own work carried out under the supervision of Mr.Devesh Pandey **and** Ms.Vaishali Dev

The matter embodied in this term-work has not been submitted by me for the award of any other degree.

Date:

…….

(Full signature of student)

PROGRAM OBJECTIVE:

1. Design a LEX Code to count the number of lines, space, tab-meta character, and rest of characters in each Input pattern.

PROGRAM CODE:

```
% {
int lines=0, spaces=0, tabs=0, others=0;
%}

%%
\n      { lines++; }
" "     { spaces++; }
\t      { tabs++; }
.       { others++; }
%%

int yywrap()  {
    return 1;
}

int main()  {
    printf("Enter input (Ctrl+D to end):\n");
    yylex();
    printf("Lines: %d\n", lines);
    printf("Spaces: %d\n", spaces);
    printf("Tabs: %d\n", tabs);
    printf("Other characters: %d\n", others);
    return 0;
}
```

OUTPUT:-

```
----- Q1: Count Lines, Spaces, Tabs, Others -----
Enter input (Ctrl+D to end):
Lines: 3
Spaces: 3
Tabs: 1
Other characters: 26
```

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PROGRAM OBJECTIVE:

2. **Design a LEX Code to identify and print valid Identifier of C/C++ in given Input pattern.**

PROGRAM CODE:

```
% {
#include <stdio.h>
%}

%%
[a-zA-Z_][a-zA-Z0-9_]*   { printf("Valid Identifier: %s\n", yytext); }
[0-9][a-zA-Z0-9_]*       { printf("Invalid Identifier: %s\n", yytext); }
\n                       { }
.                        { }
%%

int yywrap()  {
   return 1;
}

int main()  {
   printf("Enter input (Ctrl+D to end):\n");
   yylex();
   return 0;
}
```

OUTPUT:-

```
----- Q2: Valid C/C++ Identifiers -----
Enter input (Ctrl+D to end):
Valid Identifier: myVar
Valid Identifier: _count
Invalid Identifier: 2ndValue
Valid Identifier: name1
Valid Identifier: _temp
Invalid Identifier: 123abc
Valid Identifier: int
Valid Identifier: hello_world
```

PROGRAM OBJECTIVE:

3. **Design a LEX Code to identify and print integer and float value in given Input pattern.**

PROGRAM CODE:

```
% {
#include <stdio.h>
%}

%%
[0-9]+\.[0-9]+    { printf("Float: %s\n", yytext); }
[0-9]+            { printf("Integer: %s\n", yytext); }
\n                { }
.                 { }
%%

int yywrap()  {
   return 1;
}

int main()  {
   printf("Enter input (Ctrl+D to end):\n");
   yylex();
   return 0;
}
```

OUTPUT:-

```
----- Q3: Integer and Float Values -----
Enter input (Ctrl+D to end):
Integer: 42
Float: 3.14
Integer: 100
Float: 0.5
Integer: 7
Float: 99.99
Float: 12.0
```

PROGRAM OBJECTIVE:

4. **Design a LEX Code for Tokenizing (Identify and print OPERATORS, SEPERATORS, KEYWORDS, IDENTIFERS) in the C-fragment:**

PROGRAM CODE:

```
% {
#include <stdio.h>
%}

%%
"int"|"float"|"char"|"double"|"if"|"else"|"while"|"for"|"do"|"return"|"void"|"main"|"include"|"pr
intf"|"scanf"   { printf("Keyword: %s\n", yytext); }

"+"|"-"|"*"|"/"|"%"|"="|"=="|"!="|"<"|">"|"<="|">="|"&&"|""|"!"   { printf("Operator: %s\n",
yytext); }

";"|","|"("|")"|"{"|"}"|"["|"]"|"#"   { printf("Separator: %s\n", yytext); }

[a-zA-Z_][a-zA-Z0-9_]*   { printf("Identifier: %s\n", yytext); }

[0-9]+(\.[0-9]+)?        { printf("Number: %s\n", yytext); }

\"[^\"]*\"               { printf("String: %s\n", yytext); }

\n                       { }
[ \t]+                   { }
.                        { }

%%

int yywrap()  {
   return 1;
}

int main()  {
   printf("Enter C code (Ctrl+D to end):\n");
   yylex();
   return 0;
}
```

OUTPUT: -

```
----- Q4: Tokenizing C Code Fragment -----
Enter C code (Ctrl+D to end):
Separator: #
Keyword: include
Keyword: int
Keyword: main
Separator: (
Separator: )
Separator: {
Keyword: int
Identifier: x
Operator: =
Number: 10
Separator: ;
Keyword: float
Identifier: y
Operator: =
Number: 3.14
Separator: ;
Keyword: if
Separator: (
Identifier: x
Operator: >
Number: 5
Separator: )
Separator: {
Keyword: printf
Separator: (
String: "hello"
Separator: )
Separator: ;
Separator: }
Keyword: return
Number: 0
Separator: ;
Separator: }
```

PROGRAM OBJECTIVE:

5.Design a LEX Code to count and print the number of total characters, words, white spaces in given 'Input.txt' file.

PROGRAM CODE:

```
% {
int chars=0, words=0, spaces=0;
%}

%%
[a-zA-Z0-9]+   { words++; chars += yyleng; }
" "          { spaces++; chars++; }
\t           { spaces++; chars++; }
\n           { chars++; }
.           { chars++; }
%%

int yywrap()  {
   return 1;
}

int main()  {
   FILE *fp = fopen("Input.txt", "r");
   if(fp == NULL)  {
      printf("Cannot open Input.txt\n");
      return 1;
   }
   yyin = fp;
   yylex();
   fclose(fp);
   printf("Characters: %d\n", chars);
   printf("Words: %d\n", words);
   printf("White Spaces: %d\n", spaces);
   return 0;
}
```

OUTPUT: -

```
----- Q5: Count Characters, Words, Spaces in Input.txt -----
Input.txt content:
Hello   World    this is   a test
with     multiple   spaces    and       tabs

Characters: 73
Words: 11
White Spaces: 25
```

PROGRAM OBJECTIVE:

6.**Design a LEX Code to replace white spaces of 'Input.txt' file by a single blank character into 'Output.txt' file.**

PROGRAM CODE:

```
% {
#include <stdio.h>
FILE *out;
%}

%%
[ \t]+    { fprintf(out, " "); }
\n        { fprintf(out, "\n"); }
.         { fprintf(out, "%s", yytext); }
%%

int yywrap()  {
   return 1;
}

int main()  {
   FILE *fp = fopen("Input.txt", "r");
   if(fp == NULL)  {
      printf("Cannot open Input.txt\n");
      return 1;
   }
   out = fopen("Output.txt", "w");
   if(out == NULL)  {
      printf("Cannot create Output.txt\n");
      fclose(fp);
      return 1;
   }
   yyin = fp;
   yylex();
   fclose(fp);
   fclose(out);
   printf("Done! Check Output.txt\n");
   return 0;
}
```

OUTPUT: -

```
----- Q6: Replace Whitespaces -> Output.txt -----
Done! Check Output.txt
Output.txt content:
Hello World this is a test
with multiple spaces and tabs
```

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